

Stress (Structural) Pass via Submodel Temperature Mapping — Orphan-Mesh- Modified Cooldown

Goal: Run the structural half of the sequentially-coupled stress-relief cooldown on a **finer, denser full-assembly mesh**, loaded by the temperature field computed in the coarse thermal pass. Because the stress mesh differs from the thermal mesh, the temperature is carried across with Abaqus **node-based submodeling** (job 169), then read into a **static thermal-stress** run (job 269) that also extracts free-body section forces on the LUG.

Decisions locked for this walkthrough:

- Mesh strategy: **finer submodel + submodel temperature mapping** (the 169/269 route).
- Reference / initial temperature: **keep 70 °F** as in the deck (see §3.3 for exactly when that is legitimate and what must stay consistent).
- **Single governing time**, $t = 78,015.2$ s.
- Scope: linear elastic, statically-determinate restraint, **LUG section-force extraction** kept.

*Terminology note: "submodel" here does **not** mean a small local cut-out. Your stress model covers the **entire assembly**; submodeling is used only as the temperature-mapping tool between the coarse thermal mesh and the fine stress mesh. Job 269 is a full, standalone stress model — it is not driven by displacements from any global stress model (there isn't one; the global pass was thermal-only).*

1. The two-job architecture and data flow

```

Global THERMAL run (coarse mesh, §10 deck)
    |
    | writes NT(x, t) transient history
    ▼
thermal.odb —————
    |
    | submodel interpolation (spatial), DOF 11
    ▼
JOB 169 *SUBMODEL + *BOUNDARY,SUBMODEL ← picks the single instant t = 78015.2 s
(fine mesh, heat-transfer element types)
    |
    | writes NT(x) on the FINE mesh
    ▼
169.fil / 169.odb —————
    |
    | node-to-node read (same fine mesh) *TEMPERATURE, FILE=169
    ▼
JOB 269 *STATIC + *TEMPERATURE ← thermal stress on the fine mesh
(same fine mesh, structural element types)
    |
    | — S, TRES (stress intensity) , U, NT → .odb/.dat/.fil
    | — *SECTION PRINT on LUG → SOAREA / SOF / SOM / SOCF

```

Two mesh relationships, and this is the key to keeping error low:

1. **Global (coarse) → 169 (fine):** meshes differ, so Abaqus **interpolates** the temperature. This is the only interpolation step, and per §8.2 interpolation of a temperature field across dissimilar meshes near a gradient is where error creeps in — so treat job 169 as the accuracy-critical step.
2. **169 (fine HT) → 269 (fine structural):** the **same fine mesh**, only the element types change (heat-transfer → structural). Temperature transfers **node-to-node with zero interpolation error** — the clean §8.2 case. Build one fine mesh and reuse it for both jobs.

2. Prerequisites from the thermal pass

Before starting, confirm:

- The **global thermal .odb** (the orphan-mesh-modified §10 run you completed) contains **NT written at or bracketing t = 78,015.2 s**. Submodel driving reads from output database time points; if the field wasn't written near that time, the interpolation in time is coarse. (This is the §10.8 "output points near the peak-ΔT time" note.)
- You have identified **78,015.2 s as the governing instant** — the time whose combination of stress and temperature-dependent yield gives the worst stress ratio (§2.3 warns this is usually mid-cooldown and not obvious; if you only inferred it, re-check against the thermal history).
- The **fine stress mesh is built on the AS-MODIFIED geometry** — i.e. it must include the swapped/modified dome region, not the original shape. The submodel interpolation is purely spatial, so the fine mesh geometry has to match the real (modified) part or the mapped temperatures land on the wrong material.

3. Concepts you need before writing the decks

3.1 Node-based submodeling for temperature (job 169)

*SUBMODEL declares that the driven nodes get their prescribed values by **interpolating the global .odb** at their spatial location. *BOUNDARY, SUBMODEL, STEP=1 with DOF 11, 11 prescribes **temperature** (DOF 11) on the driven node set. STEP=1 selects which step of the global analysis to read.

- **Which nodes to drive — MODEL.** For a whole-assembly temperature remap, drive **every node** (make MODEL contain all nodes). Then every fine node gets its interpolated temperature directly, and the steady-state heat-transfer step is fully prescribed — it simply writes the mapped field out. If you drive **only boundary nodes** and let *HEAT TRANSFER, STEADY STATE fill the interior, you get the *steady conduction* solution consistent with those boundary temperatures — which is **not** the true transient interior field during a cooldown. For a cooldown, the interior is not quasi-steady, so **drive all nodes** unless you have a specific reason not to.
- **EXTERIOR TOLERANCE.** Your deck uses 0.25 (25 % of the local global element size); the default is 0.05. A fine, curved submodel surface often has driven nodes that fall slightly *outside* the faceted coarse global mesh, and the tolerance is how far outside a node may sit before Abaqus errors instead of extrapolating. 0.25 is loose but defensible near a curved dome; the trade-off is that too loose a tolerance lets a stray node grab a distant global element and interpolate a wrong temperature. Check the .dat for the list of driven nodes and their exterior distances, and tighten if any are being pulled from far away.
- **Linking the global model.** The global results are named at submission, not in the deck: `abaqus job=169`
`globalmodel=<thermal_job_name> input=169.inp interactive.` Optionally add `GLOBAL ELSET= on`
`*SUBMODEL` to restrict the search to part of the global mesh (useful to avoid grabbing the wrong side across a thin tie gap — see §5).

3.2 Reading the mapped field into the stress run (job 269)

*TEMPERATURE, FILE=169, BSTEP=1, ESTEP=1 reads NT from **job 169's results file** (169.fil — your FILE=169.inp is a misnomer; it is the job/results name, not the input file). BSTEP/ESTEP select the step range. Because 269 uses the **same fine mesh** as 169, the read is node-to-node and needs **no INTERPOLATE** parameter. Submit with `oldjob=169` so the read finds the file: `abaqus job=269 oldjob=169 input=269.inp interactive.`

3.3 Reference / initial temperature — keeping 70 °F correctly

Your deck sets *INITIAL CONDITIONS, TYPE=TEMPERATURE, ALL, 70, while §6.2 argues the strain-free reference should be the **1100 °F soak**. Both can be correct; here is the precise reasoning so keeping 70 °F is a deliberate, defensible choice rather than an oversight:

- Abaqus computes thermal strain as $\epsilon_{th} = \alpha(T) \cdot (T - \theta^0)$, where θ^0 is the ***EXPANSION reference (ZERO=)**, *not* the initial-condition temperature. The *INITIAL CONDITIONS value only sets where the step starts; for a single static step the **end-state** stress depends on the applied field, θ^0 , $\alpha(T)$, and the restraint — not on the path from 70 °F.
- In a **statically-determinate (free) structure**, stress is produced only by the **spatially-varying** part of ϵ_{th} . Adding a constant to θ^0 (or shifting all temperatures by a constant) changes ϵ_{th} by a uniform amount, which is pure stress-free

expansion. So with a true 3-2-1 restraint, **70 °F and 1100 °F references give the same stresses** — 70 °F is fine.

- The choice **stops being stress-neutral the moment any restraint carries uniform-contraction reaction**. If your supports are not statically determinate, the reference matters and §6.2's 1100 °F becomes the correct one. So the validity of "keep 70 °F" is tied directly to the restraint being genuinely 3-2-1 (§3.4). Verify the support reactions are ~0 (§6).
- **The one thing that must stay consistent:** the `*EXPANSION` table's secant α values are referenced to a specific θ^0 (§6.1). If you keep 70 °F, set `*EXPANSION, ZERO=70` and make sure the α table is the secant curve referenced to 70 °F. Mixing an α table referenced to one temperature with a different `ZERO=` silently produces thermal-strain errors of several percent that map straight into stress.

Bottom line: keeping 70 °F is legitimate *provided* the restraint is statically determinate and `*EXPANSION, ZERO=` matches the α table's reference. If you cannot guarantee determinate restraint, switch to 1100 °F per §6.2.

3.4 Restraint — the highest-risk item (§8.3)

There is **no applied mechanical load**; every pound of support reaction is fictitious stress. Use a **statically-determinate 3-2-1** restraint that removes the six rigid-body modes and nothing more:

- Pick three well-separated nodes on stiff regions. Node A: fix DOF 1,2,3. Node B: fix 2,3 (frees 1). Node C: fix 3 (frees 1,2). Adapt the axes to your geometry so the structure can contract freely in every direction.
- After the run, confirm the **reaction forces at A/B/C are essentially zero** — a nonzero reaction means you have injected stress.

3.5 Stress measure and the LUG section extraction

- Output **TRESC (Tresca = stress intensity)** alongside `S`. ASME-lineage limits are written in stress intensity; reporting von Mises against a stress-intensity limit is non-conservative by up to 15.5 % (§2.3). Your deck already requests `S, TRESC` — keep it.
- ***SECTION PRINT free-body cut on LUG** integrates the stresses over the `LUG` surface and reports the **net section area/force/moment/center** (`SOAREA/SOF/SOM/SOCF`) in a **local axis system** defined by the coordinate data lines. This is your lug (or weld) load for downstream hand calcs. `AXES=LOCAL` (your `AXEX` is a typo); `FREQ=99` just limits how often it prints.

4. Job 169 — corrected, annotated skeleton

```

** JOB 169 - map global thermal field onto the fine assembly mesh @ t = 78015.2 s
** Submit: abaqus job=169 globalmodel=<thermal_job_name> input=169.inp interactive
*HEADING
Temperature map onto fine assembly mesh @ t=78015.2 s
**
** ---- FINE mesh, heat-transfer element types (same family as the global thermal deck) ----
*NODE, NSET=ALL
    <fine mesh nodes>
*ELEMENT, TYPE=DC3D8, ELSET=SOLIDS
    <...>
*ELEMENT, TYPE=DS4, ELSET=SHELLS
    <...>
**
** Driven set: ALL nodes -> full-field temperature map (see 3.1)
*NSET, NSET=MODEL
    ALL
**
*SOLID SECTION, ELSET=SOLIDS, MATERIAL=THERMAL_MAT
*SHELL SECTION, ELSET=SHELLS, MATERIAL=THERMAL_MAT
    <thickness>, <n_pts>
*MATERIAL, NAME=THERMAL_MAT
*CONDUCTIVITY
    <k(T)>
*SPECIFIC HEAT
    <cp(T)>
*DENSITY
    <rho>
**
** ---- Submodel declaration: source is the global thermal .odb (named at submission) ----
*SUBMODEL, EXTERIOR TOLERANCE=0.25
    MODEL
**
*STEP, INC=1
*HEAT TRANSFER, STEADY STATE
    78015.2, 78015.2
** Prescribe DOF 11 (temperature) on the driven nodes by interpolation from the global run
*BOUNDARY, SUBMODEL, STEP=1
    MODEL, 11, 11
*NODE FILE
    NT
*OUTPUT, FIELD, FREQ=1
*NODE OUTPUT
    NT
*END STEP

```

Corrections vs your snippet: `INTERIOR` → `EXTERIOR` `TOLERANCE`; add the `MODEL` node set and material/section blocks (elided in your snippet); confirm the global model is named with `globalmodel=` at submission.

5. Job 269 — corrected, annotated skeleton

```

** JOB 269 - static thermal-stress on the fine mesh; reads mapped T from job 169
** Submit: abaqus job=269 oldjob=169 input=269.inp interactive
*HEADING
Cooldown thermal stress @ t=78015.2 s
**
** ---- SAME fine mesh as 169, STRUCTURAL element types ----
*NODE, NSET=ALL
    <identical fine mesh nodes as 169>
*ELEMENT, TYPE=C3D8, ELSET=SOLIDS
    <identical connectivity, structural type>
*ELEMENT, TYPE=S4, ELSET=SHELLS
    <...>
**
*SOLID SECTION, ELSET=SOLIDS, MATERIAL=STEEL, ORIENTATION=CYL
*SHELL SECTION, ELSET=SHELLS, MATERIAL=STEEL
    <thickness>, <n_pts>
*ORIENTATION, NAME=CYL, SYSTEM=CYLINDRICAL
    <a1,a2,a3, b1,b2,b3>
**
*MATERIAL, NAME=STEEL
*ELASTIC
    <E(T)>, <nu(T)>, <T>
*EXPANSION, ZERO=70
    <alpha_secant(T) referenced to 70 F>, <T>
**
** ---- Surface for the lug free-body cut ----
*SURFACE, TYPE=ELEMENT, NAME=LUG
    <lug_elset>, <Sx face>
**
** ---- 3-2-1 statically determinate restraint (rigid-body only) ----
*NSET, NSET=RB_A
    <node A>
*NSET, NSET=RB_B
    <node B>
*NSET, NSET=RB_C
    <node C>
**
*INITIAL CONDITIONS, TYPE=TEMPERATURE
    ALL, 70
**
*STEP
*STATIC
** read the mapped temperature field from job 169 (node-to-node, same mesh)
*TEMPERATURE, FILE=169, BSTEP=1, ESTEP=1
**

```

```

*BOUNDARY
  RB_A, 1, 3
  RB_B, 2, 3
  RB_C, 3, 3
**
*SECTION PRINT, NAME=LUG, SURFACE=LUG, AXES=LOCAL, FREQ=99
  <xa, ya, za>          ** local-axis definition point(s) for your lug geometry
  <xb, yb, zb, ...>
SOAREA
SOF
SOM
SOCF
**
*FILE FORMAT, ASCII
*NODE PRINT
  U, NT
*NODE FILE, GLOBAL=YES
  U, NT
*EL PRINT, POSITION=AVERAGED
  S, TRES
*EL FILE, POSITION=NODS
  S
*OUTPUT, FIELD, FREQ=1
*NODE OUTPUT
  U, NT
*ELEMENT OUTPUT
  S
*END STEP

```

Corrections vs your snippet: FILE=169.inp → FILE=169 (results of job 169, 169.fil); AXEX → AXES; POS=AVERAGE → POSITION=AVERAGED, POS=NODS → POSITION=NODS (abbreviations usually accept, but use the full forms); add the ***ELASTIC / *EXPANSION, ZERO=70** material, the ***ORIENTATION** (cylindrical, since the thermal deck used TYPE=ORTHO conductivity — the structural pass needs the consistent frame), and the **3-2-1 *BOUNDARY** block that your snippet omitted.

6. Running order and file dependencies

1. Global thermal run → thermal.odb (already done).
2. abaqus job=169 globalmodel=<thermal_job> input=169.inp interactive → 169.fil, 169.odb.
3. abaqus job=269 oldjob=169 input=269.inp interactive → 269.odb, 269.dat (section prints), 269.fil.

oldjob=169 is what lets 269's *TEMPERATURE, FILE=169 find the file. If you rename jobs, keep the FILE= value and oldjob= in sync.

7. Orphan-mesh-modified specifics

Because your thermal model was built by the orphan-mesh region-swap (modified region tied in at labels 20001+), watch these in the stress pass:

- **The fine mesh must reflect the modified geometry (§2).** Mesh the as-modified dome, not the original.
- **Interpolation across the tie interface.** Submodel mapping is spatial, so it does not care about the thermal model's tie or the 20001+ labels — but if the tie left a small physical gap or overlap, fine driven nodes near that seam can fall outside the coarse global elements. That is exactly what `EXTERIOR TOLERANCE=0.25` is buying you; still, check the `.dat` driven-node report near the modified region and, if needed, use `GLOBAL ELSET=` to keep the search on the correct side of the seam.
- **Temperature continuity.** The thermal tie enforced temperature continuity across the seam, so the global field you are sampling should be smooth there. If you see a kink in the mapped `NT` at the seam, suspect the thermal tie or a mapping tolerance issue before trusting the stress.
- **Material and section on the modified region.** The new region's elset needs `*ELASTIC + *EXPANSION` and the cylindrical `*ORIENTATION` assigned in 269, exactly as tracked in the change-control ledger for the thermal pass. An unassigned modified region silently drops out of the stress result.

8. Verification checklist

Run these before trusting the numbers:

1. **Mapping sanity (169).** Compare min/max `NT` on the fine mesh at $t = 78015.2$ s against the global thermal field at that time — they should match within interpolation error. A fine-mesh extreme well outside the global range means a bad exterior-tolerance pick.
2. **No dropped driven nodes.** The `169.dat` should list every driven node as successfully mapped; investigate any warned/extrapolated nodes near the modified seam.
3. **Node-to-node read (269).** Confirm `*TEMPERATURE, FILE=169` loaded (check `269.dat / .msg`); `NT` echoed in 269 should equal the 169 field node-for-node (same mesh).
4. **Restraint reactions ~0 (269).** Sum the reaction forces at `RB_A/B/C`; near-zero confirms the 3-2-1 is not injecting stress — and confirms the 70 °F reference is stress-neutral (§3.3).
5. **Reference consistency.** `*EXPANSION, ZERO=` matches the α table's reference temperature.
6. **Stress measure.** Report against 1.3-Sy using `TRESC` (stress intensity), evaluated per node at that node's temperature (`NT`). Whether to use peak or linearized stress against the limit is the open §2.3 question — decide before reporting.

If the elastic criterion is exceeded, remember it is a secondary, strain-controlled stress (§2.2): the elastic model over-predicts, and escalation is a separate elastic-plastic (or creep-relaxation) analysis, not a reinterpretation of this result.

9. Quick reference — what each keyword is doing

Keyword	Job	Purpose
*SUBMODEL, EXTERIOR TOLERANCE	169	Declare submodel; source = global .odb; how far a driven node may sit outside the global mesh
*BOUNDARY, SUBMODEL, STEP=1 ... 11,11	169	Prescribe temperature (DOF 11) on driven nodes by interpolation
*HEAT TRANSFER, STEADY STATE t,t	169	Vehicle to evaluate/write the mapped field at time t
*NODE FILE NT	169	Write fine-mesh temperature to 169.fil for the stress job
*TEMPERATURE, FILE=169, BSTEP/ESTEP	269	Read the mapped field as the predefined temperature load
*STATIC	269	Elastic thermal-stress solve
*INITIAL CONDITIONS, TYPE=TEMPERATURE	269	Starting temperature (reference handled by *EXPANSION, ZERO=)
*BOUNDARY (3-2-1)	269	Statically-determinate restraint — rigid-body only
*SECTION PRINT SOAREA/SOF/SOM/SOCF	269	Free-body net force/moment across the LUG cut, local axes
*EL PRINT/FILE S, TRESC	269	Stress tensor and Tresca (stress intensity) output